

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCEC604 - PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 In PERT analysis, the standard deviation of the overall project completion time can be estimated by
 - A) Equal to expected time
 - B) Square root of the sum of variances along the critical path
 - C) Difference between pessimistic and optimistic times
 - D) Sum of activity durations
 - 2 Which of the following options accurately describes how the value of money changes over time?
 - A) Money received in the future is worth more than the same amount received today
 - B) Money received today is worth more than the same amount received in the future
 - C) Money has the same value regardless of when it is received
 - D) Money never earns interest
 - 3 Which probability distribution is typically used to approximate the total project duration in PERT?
 - A) Beta distribution
 - B) Normal distribution
 - C) Poisson distribution
 - D) Uniform distribution
 - 4 What is the key purpose of using project management software?
 - A) Designing hardware circuits
 - B) Assigning tasks, allocating resources, and tracking project progress
 - C) Encrypting financial data only
 - D) Creating graphic animations

- 5 What is the probability that a project will be completed by its expected completion time
 - A) 0%
 - B) 50%
 - C) 100%
 - D) 25%
- 6 Computer-aided project management tools are mainly used to:
 - A) Design circuit diagrams
 - B) Plan, schedule, and monitor project activities
 - C) Create 3D models
 - D) Edit videos
- 7 Which probability distribution is normally used to model the time estimates of individual activities in PERT?
 - A) Normal distribution
 - B) Beta distribution
 - C) Gamma distribution
 - D) Exponential distribution
- 8 In project scheduling, what term describes the estimated amount of time needed to complete an activity?
 - A) Float
 - B) Duration
 - C) Slack
 - D) Lag
- 9 During the execution phase of a project, updating the CPM network is primarily done to:
 - A) Recalculate the total project budget only
 - B) Update activity durations and recalculate early/late times and critical paths
 - C) Increase activity floats deliberately
 - D) Remove critical path activities
- 10 When you crash a project to reduce its overall duration, which activities should be targeted first?
 - A) Activities with the shortest durations
 - B) Non-critical activities
 - C) Critical path activities with the lowest cost slope
 - D) Activities with the largest float
- 11 Which tool is most effective for illustrating the dependencies and sequencing of project activities?
 - A) Milestone chart
 - B) Bar (Gantt) chart without dependencies
 - C) Project network diagram (PDM)
 - D) Project budget table
- 12 A Gantt chart (bar chart) is mainly used to show:
 - A) Only the project milestones, ignoring task durations
 - B) Tasks and their durations along a timeline
 - C) The logical dependencies without any time scale
 - D) Only resource assignment details

Q2. Solve the following.

- A) Explain the differences between the Critical Path Method (CPM) and the Program Evaluation and Review Technique (PERT).

- B) Describe the different phases of Project Management. 6

Q3. Solve the following.

- A) Describe how updating calculations is done in a project network and why updating is important during project execution. 6

- B) 6

Activity	Normal Time	Crash Time	Normal Cost	Crash Cost
1-2	5	3	5000	10000
1-3	2	1	1500	3000
2-4	5	3	2000	3000
2-5	7	3	5000	8000
3-5	6	3	8000	10000
4-6	8	4	5000	6000
5-6	9	5	15000	18000

Crash the network given above and determine the Project Duration (in Days) and Project Cost.

Q4. Solve Any Two of the following.

- A) Discuss the central limit theorem in detail 6

- B) Explain the Direct Cost and indirect cost for a construction project. 6

- C) 6

Activity	Optimistic Time	Most Likely time	Pessimistic time
1-2	6	10	20
1-3	7	10	13
2-4	9	15	21
2-5	7	11	15
3-5	3	7	11
4-6	9	12	15
5-6	7	9	11

Draw the network and determine the Project Duration.

Find the Z value if the Project mentioned above has to be completed in 22 days

Q5. Solve Any Two of the following.

- A) Write a short note on Demand and Supply. 6

- B) What is the time value of money, and why is it important in financial decision-making? 6

- C) 6

Activity (i-j)	1-2	2-3	2-4	3-5	3-6	4-5	4-7	5-8	6-8	7-8
Duration (t _{ij})	5	2	6	4	4	2	3	7	8	2

Draw the network and calculate TE and TL for all activities for the following.

Q6. Solve Any Two of the following.

- A) A company manufactures a product with the following details: 6

Selling price per unit = ₹300

Variable cost per unit = ₹150

Total fixed costs = ₹4,50,000

a) Calculate the break-even point (BEP) in units.

b) Find the break-even sales revenue in rupees.

- B) Discuss how ISO standards help improve project management practices in organisations. 6

- C) Explain the concept of the Juran Trilogy and its role in improving organisational quality. 6

***** End *****